
TEXAS
WIRELESS

Do-It-Yourself (DIY)
Light-wave Sensing and Communication Project

$ Low-Cost, Portable, Effective, Impressive and Fun $

Designed by

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OUTLINE

- Demo
- Objective
- Electromagnetic Spectrum
- Modulation
- Light-wave Communication
- Instructions

OBJECTIVE

OBJECTIVE

Learn how “data” can wirelessly be transmitted using “light” by building your own

Light-wave Sensing and Communication System

Data: Audio signal

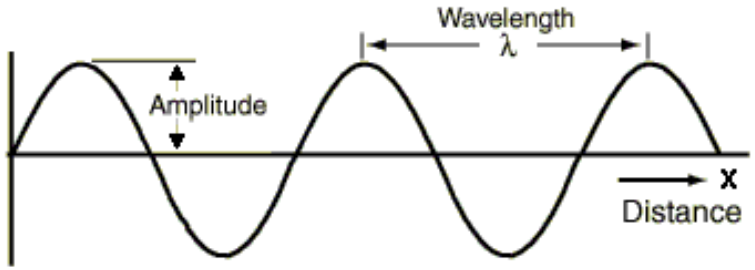
Light: Visible (White, Red, Blue, Green, Yellow) or Invisible (Infra-Red)

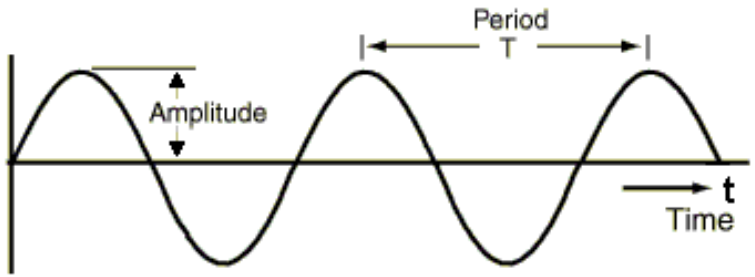
ELECTROMAGNETIC SPECTRUM

ELECTROMAGNETIC SPECTRUM

Definition (Wikipedia):

The **electromagnetic spectrum** is the range of frequencies (the spectrum) of electromagnetic radiation and their respective wavelengths and photon energies.





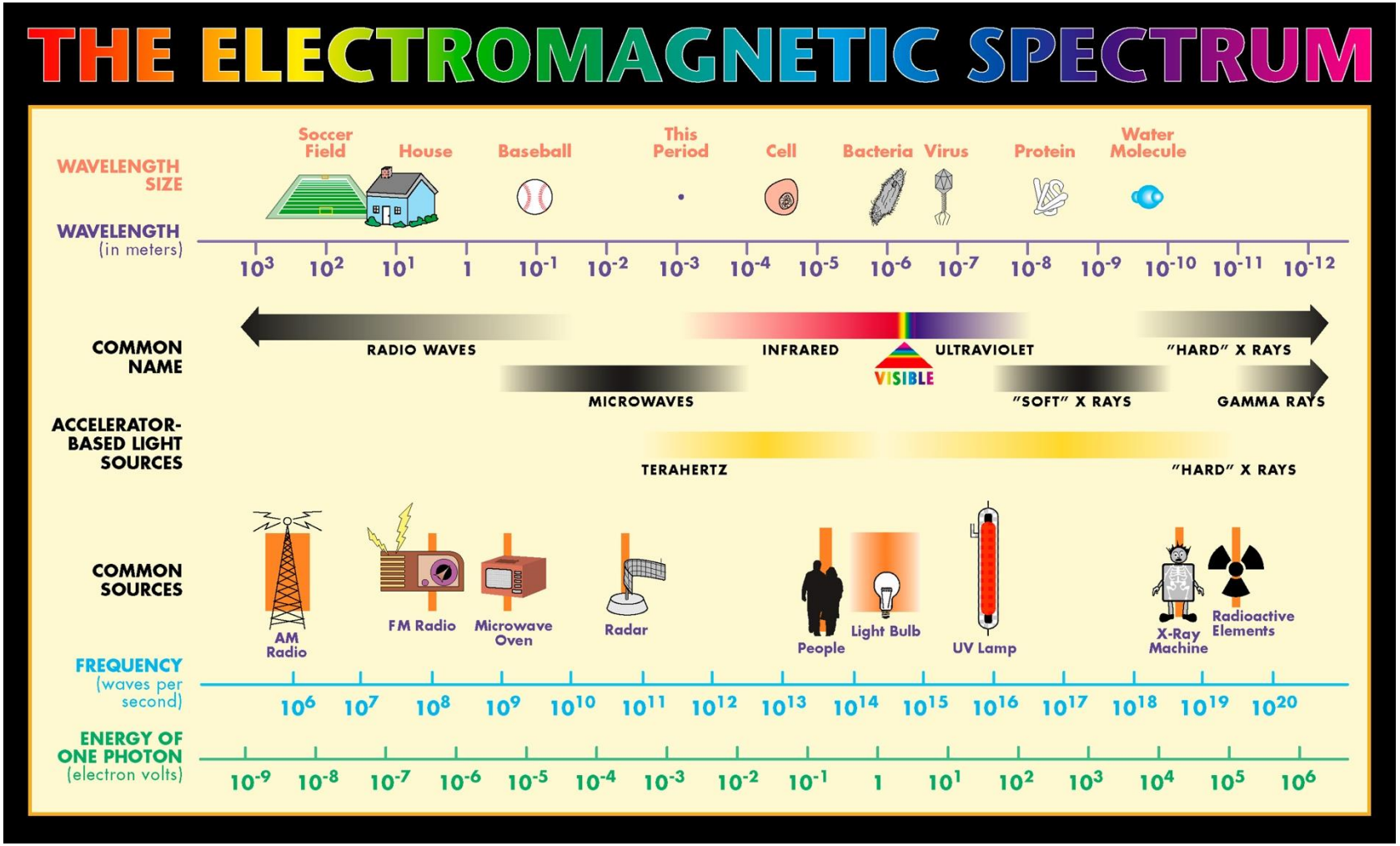
periodic waveform

Wavelength: $\lambda = \frac{c}{f}$

Frequency: $f = \frac{1}{T}$

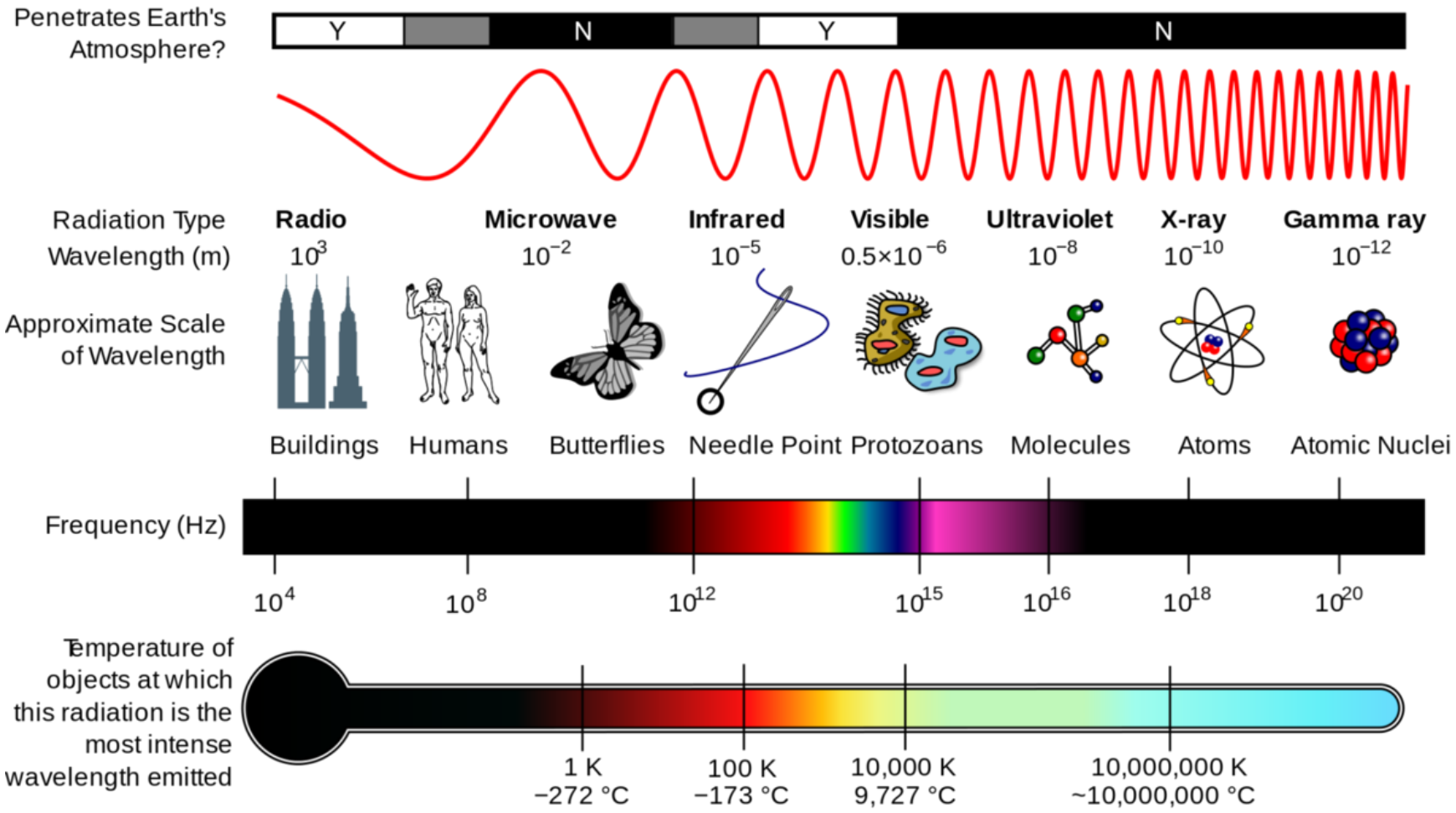
c = speed of light $\approx 3 \times 10^8$ m/s

ELECTROMAGNETIC SPECTRUM

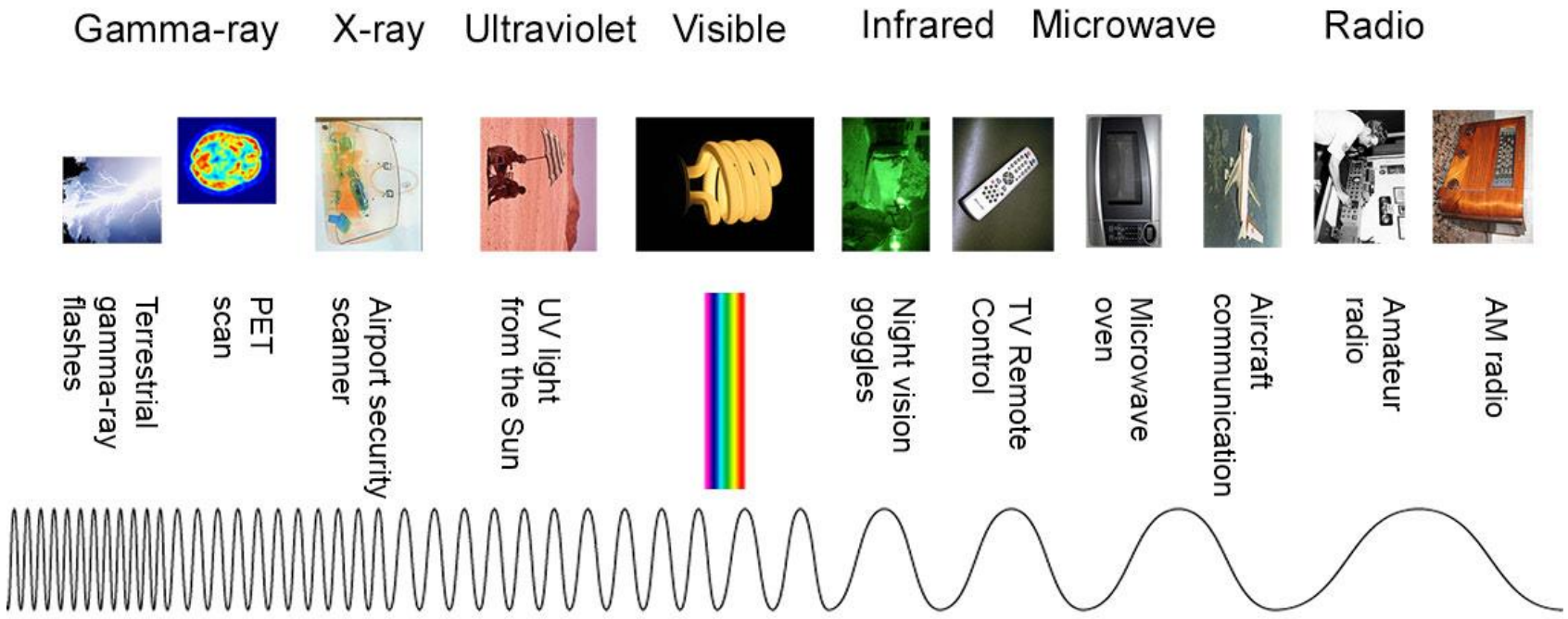


http://inclasswithdrprince.blogspot.com/2013/12/

ELECTROMAGNETIC SPECTRUM



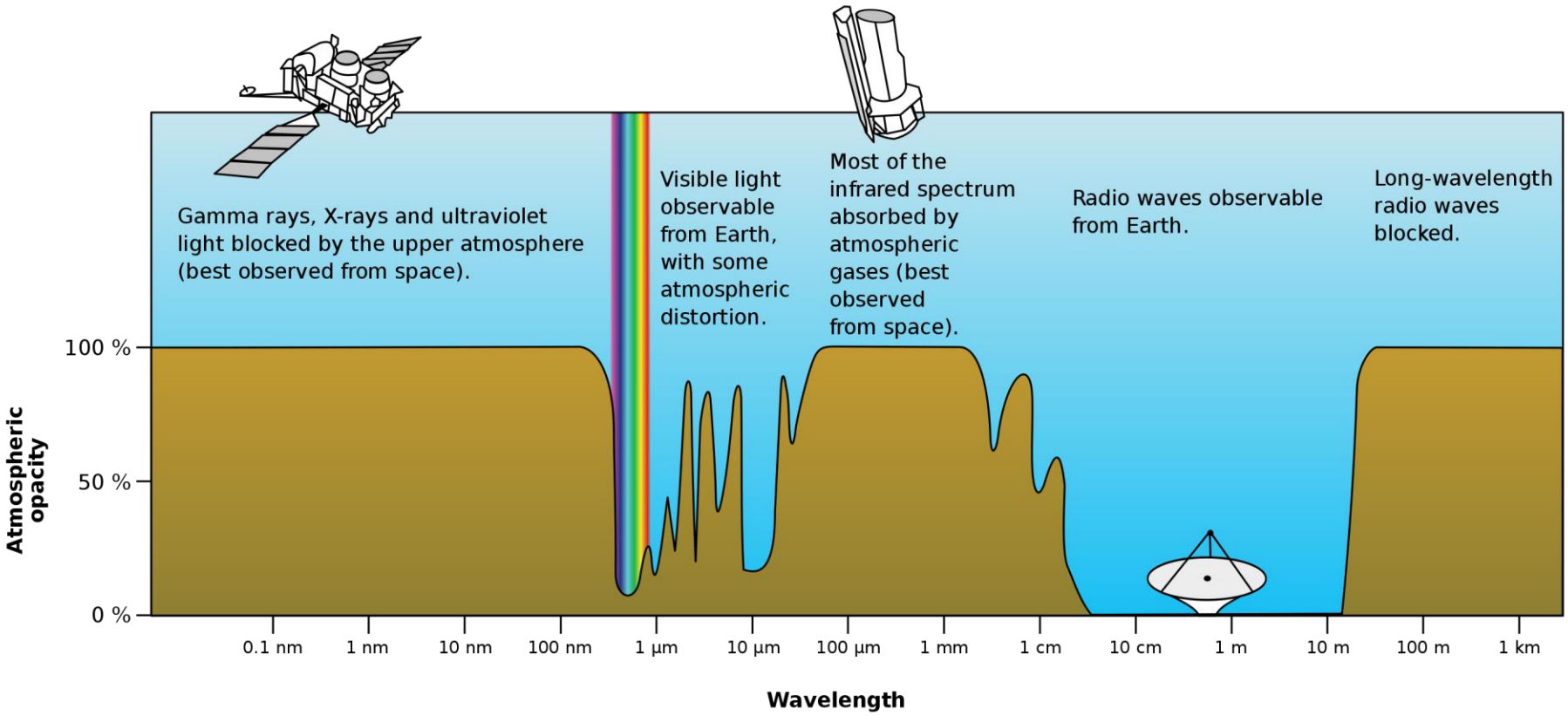
ELECTROMAGNETIC SPECTRUM



Interesting fact: "Yes, all objects, including human bodies, emit electromagnetic radiation. The wavelength of radiation emitted depends on the temperature of the objects. Such radiation is sometimes called **thermal radiation**. Most of the radiation emitted by human body is in the infrared region, mainly at the wavelength of 12 micron ($= 12 \times 10^{-6}\mathrm{m}$)".

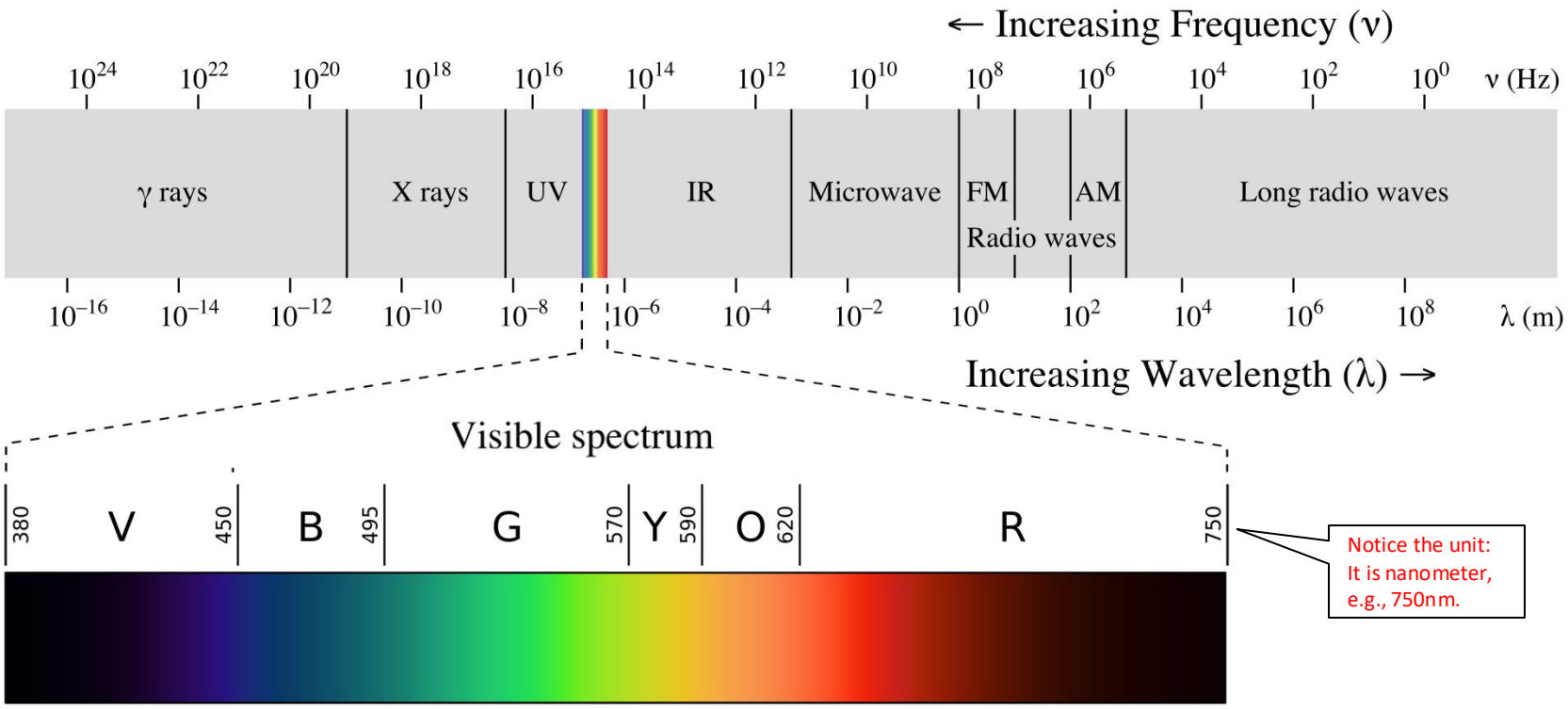
https://www.hko.gov.hk/education/edu02rga/radiation/radiation_02-e.htm

ELECTROMAGNETIC SPECTRUM

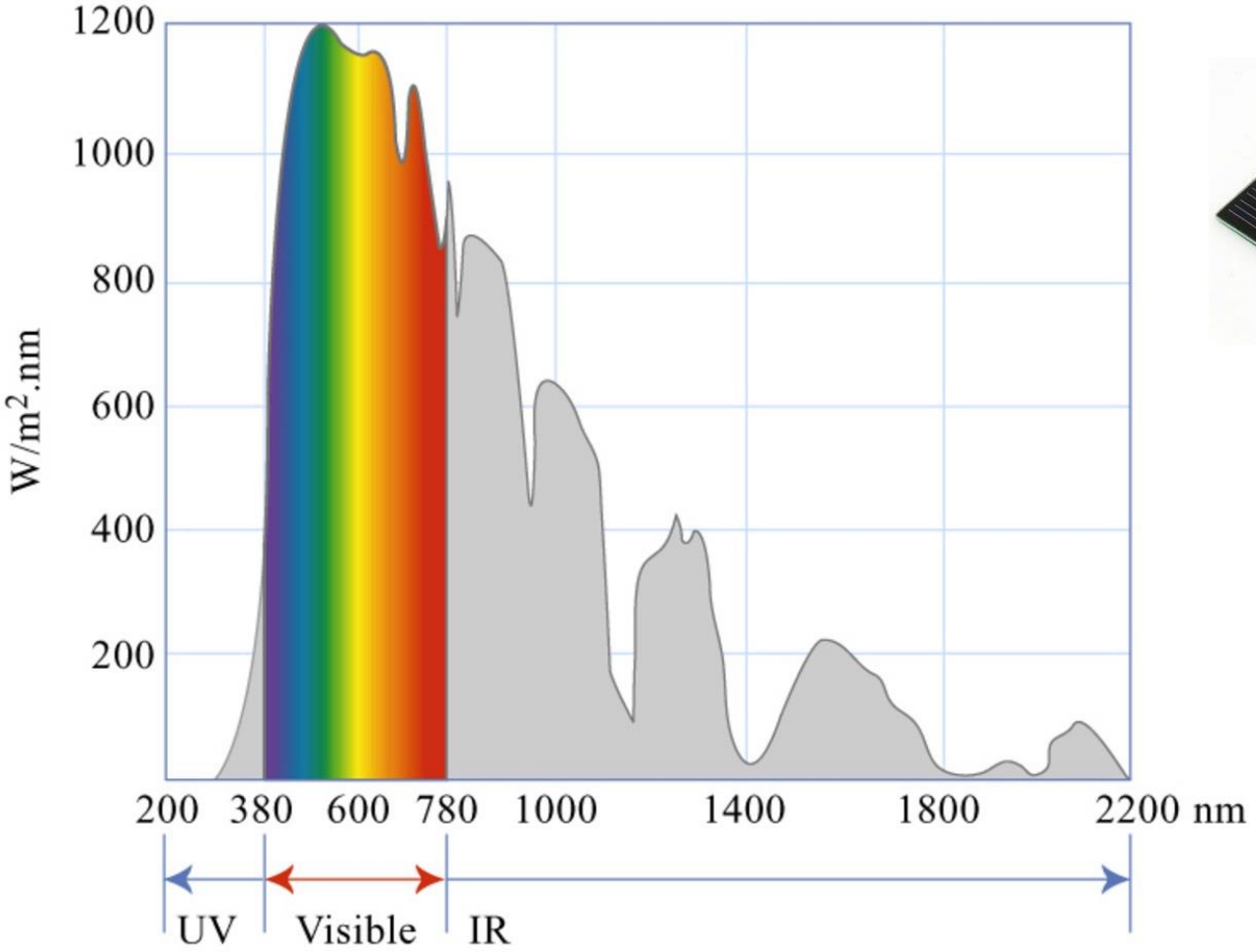


https://en.wikipedia.org/wiki/Electromagnetic_radiation

ELECTROMAGNETIC SPECTRUM



SOLAR PANEL SPECTRUM





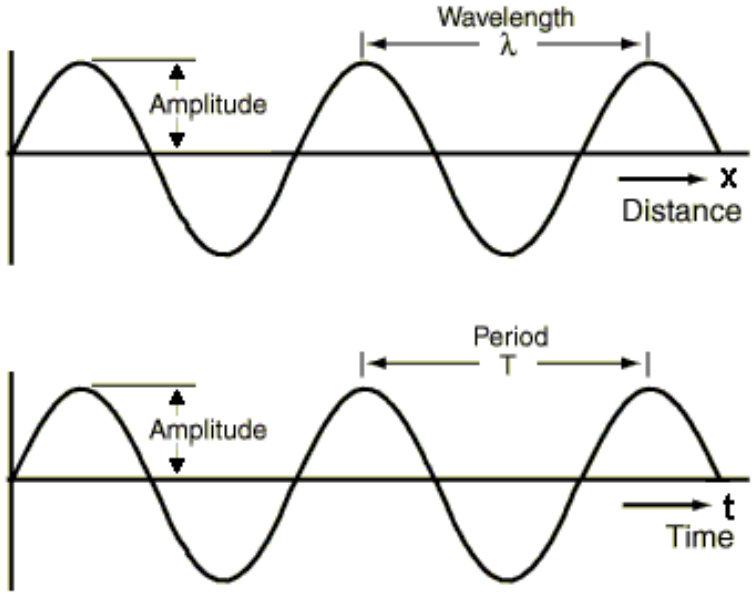
https://www.weforum.org/agenda/2015/03/how-to-make-solar-power-more-efficient-and-affordable/

MODULATION

MODULATION

Definition (Wikipedia):

In electronics and telecommunications, modulation is the process of **varying one or more properties** of a **periodic waveform**, called the carrier signal, with a modulating signal that typically contains information to be transmitted.



periodic waveform

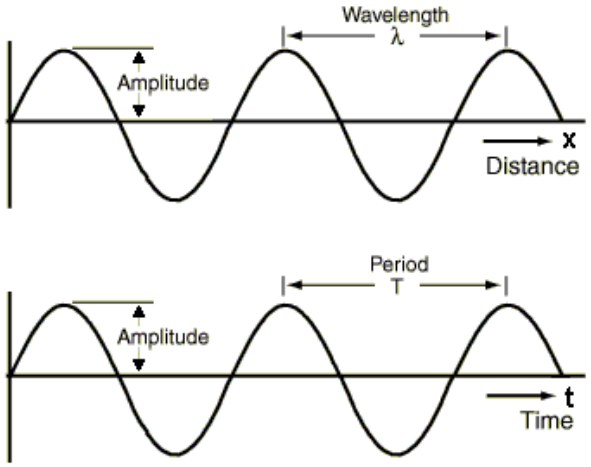
MODULATION

$$A\cos(2\pi f t + \phi)$$

Amplitude *Frequency* *Phase*

Angle

(Frequency: Rate of Change of Angle)



Frequency & Period : $f = \frac{1}{T} \leftrightarrow T = \frac{1}{f}$

Wavelength: $\lambda = \frac{c}{f} = c\left(\frac{1}{f}\right) = c\,T \;\leftrightarrow\; d = v\,t \Rightarrow (distance = speed \times time)$

$c = \text{speed of light} \approx 3 \times 10^8\ \text{m/s}$

Units: *f* (Hertz), λ (meter), *T (second)*

MODULATION

$$f = \frac{1}{T}$$

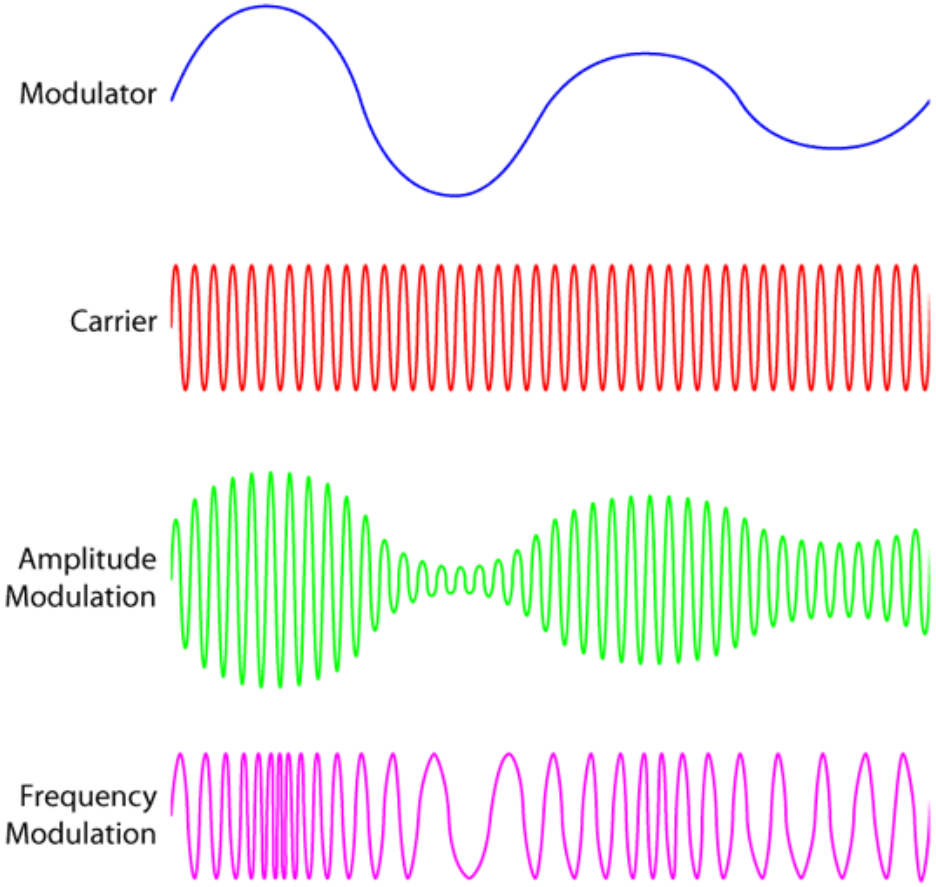
$$\lambda = \frac{c}{f} \Rightarrow f = \frac{c}{\lambda}$$

A periodic signal can be written as:

- Using Frequency: $A\cos(2\pi f t + \phi)$
- Using Period: $A\cos(2\pi(1/T)t + \phi)$
- Using Wavelength: $A\cos(2\pi(c/\lambda)t + \phi)$

Analog Modulation

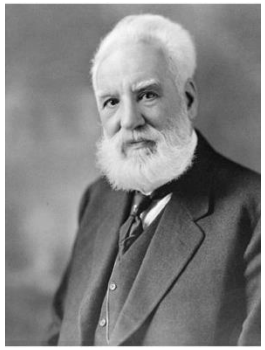
$$A\cos(2\pi f t + \phi)$$



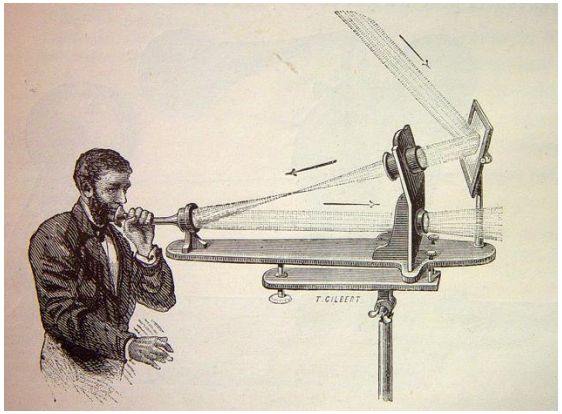
http://mriquestions.com/signal-squiggles.html

LIGHT-WAVE COMMUNICATION

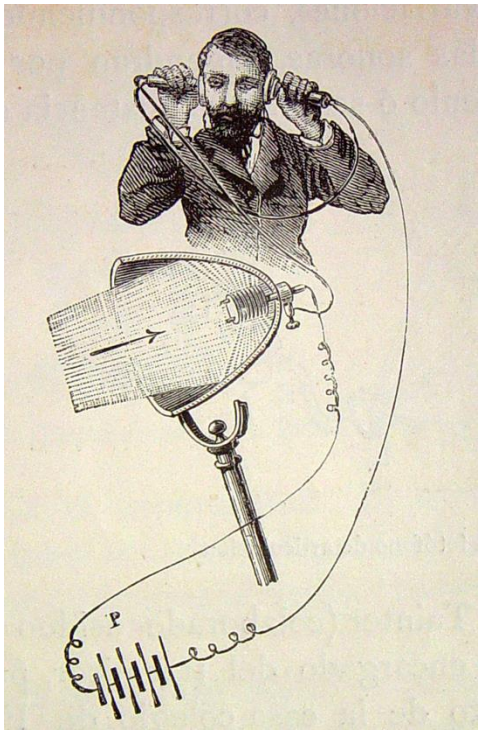
PHOTO-PHONE BY GRAHAM BELL (1880)



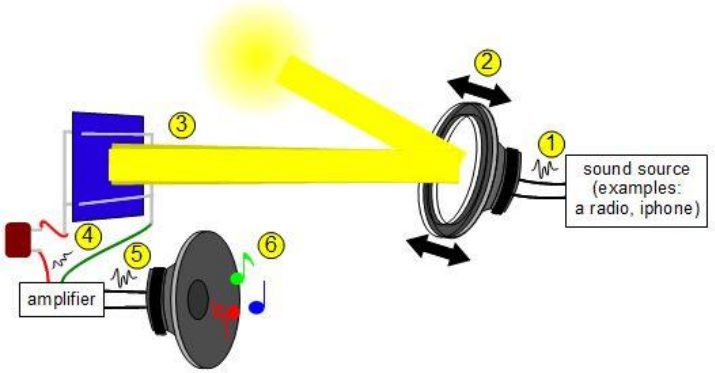
"the greatest invention [I have] ever made, greater than the telephone"



Transmitter



Receiver

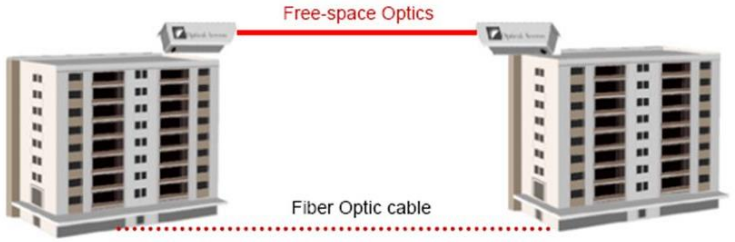


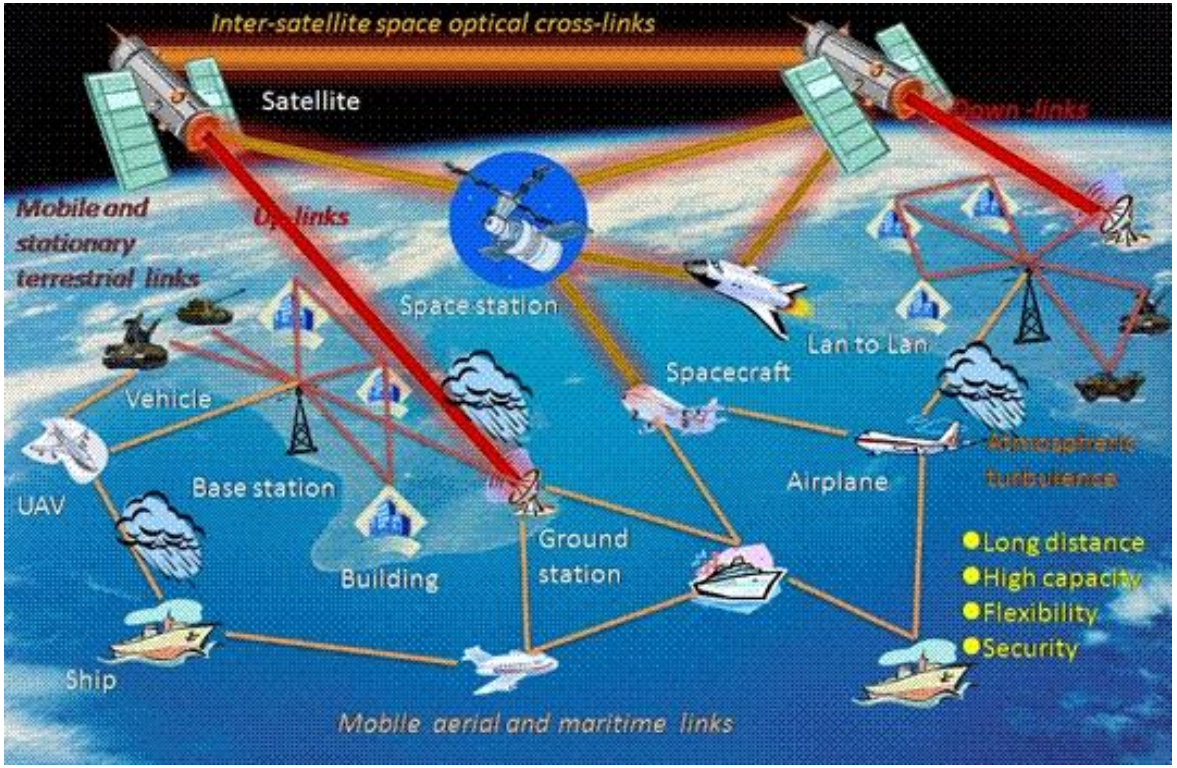
If you want to build yourself by using sunlight see the link below:
https://rimstar.org/science_electronics_projects/photophone_transmitting_sound_with_sunlight.htm

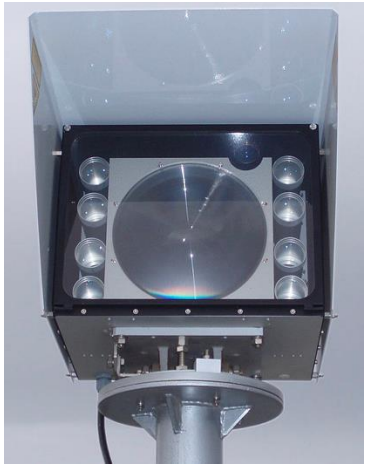
FREE-SPACE OPTICAL (FSO) COMMUNICATION

a.k.a. Laser Communication

Mostly used wavelength: 1500 and 850nm



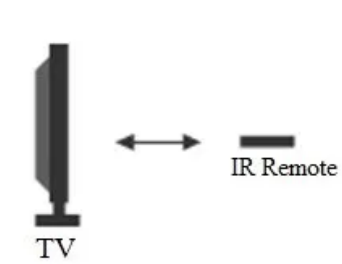




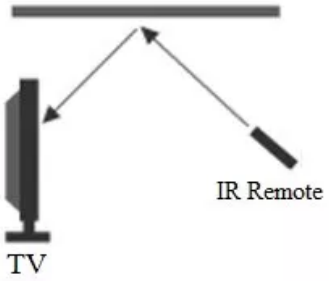
INFRA-RED (IR) COMMUNICATION



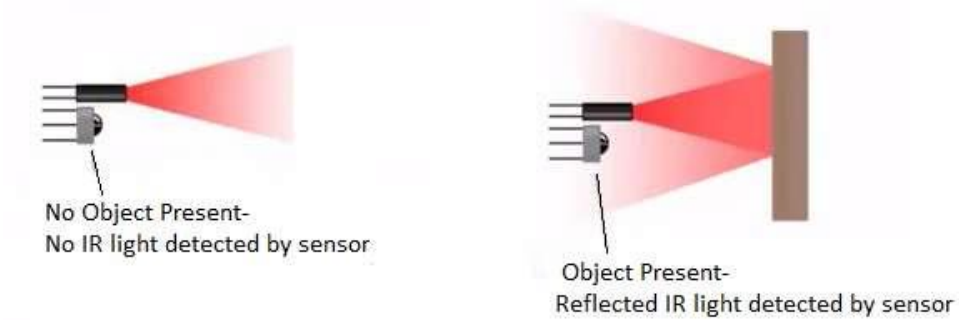




Point to Point



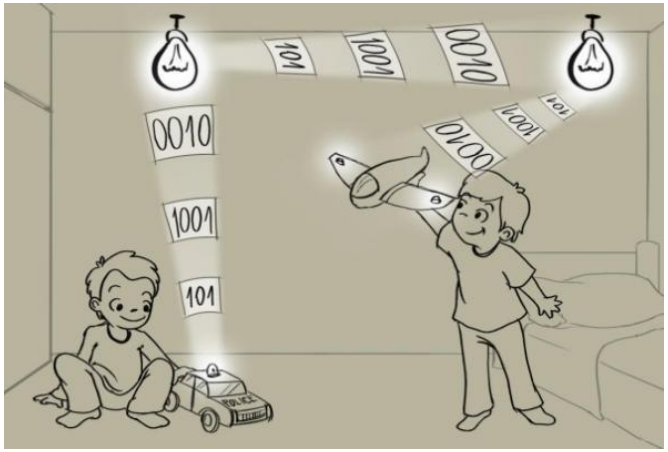
Diffuse

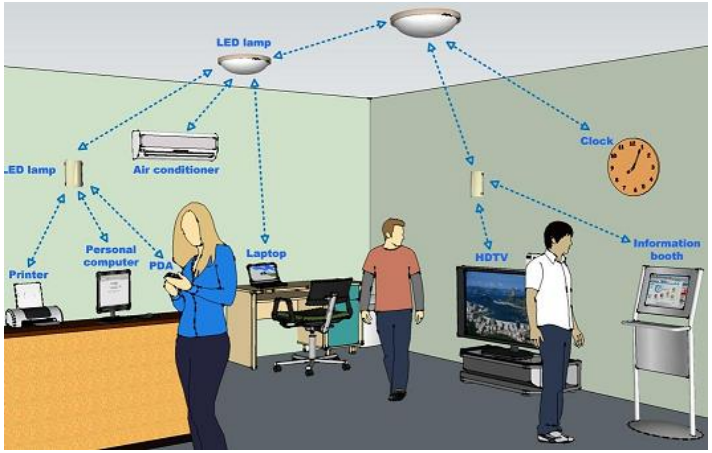


Infrasonic or Infrared Sensor

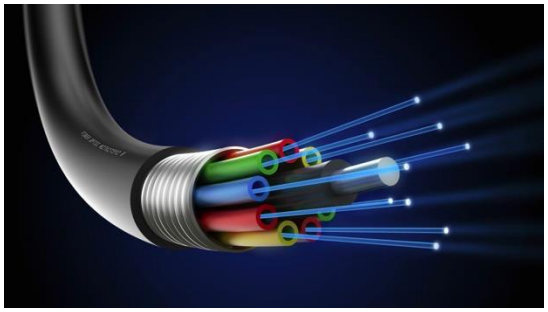
VISIBLE LIGHT COMMUNICATION



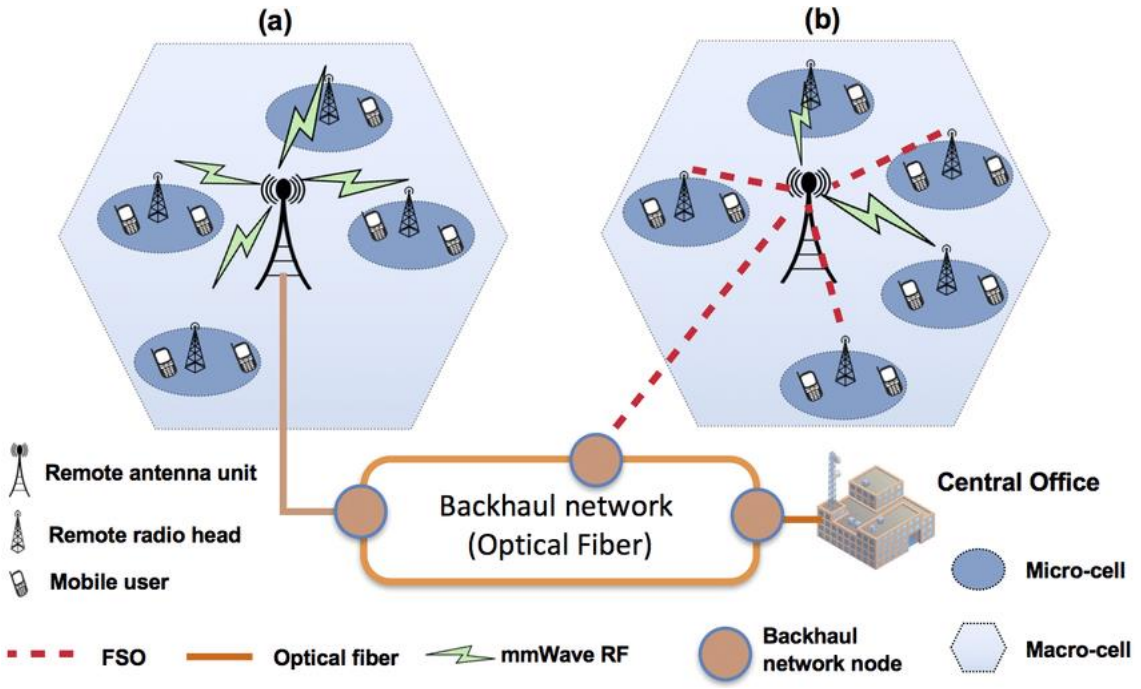




FIBER-OPTIC COMMUNICATION (WIRED)





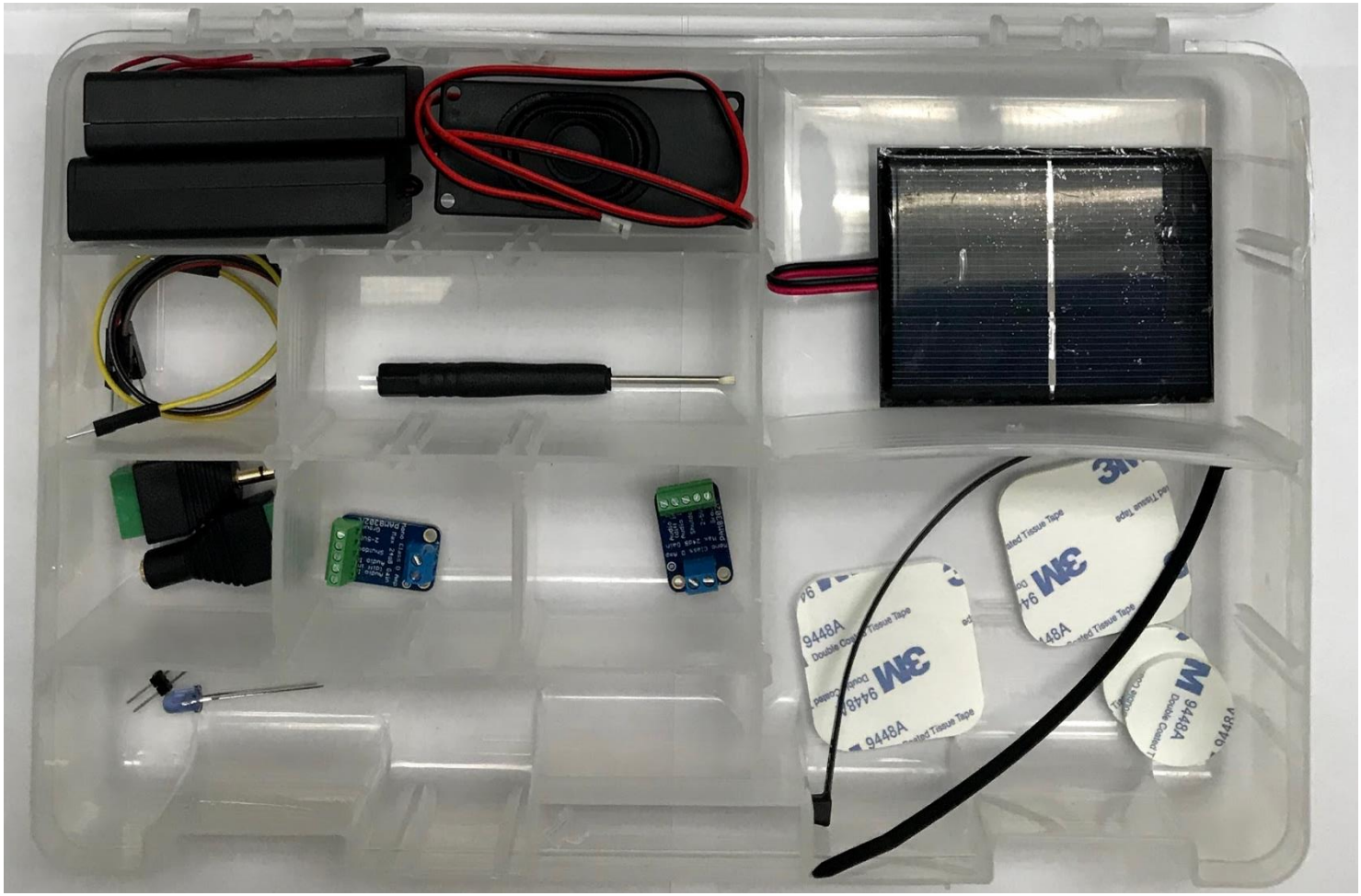


INSTRUCTIONS

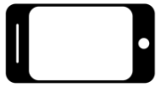
Tool needed: Screwdriver 

Have fun and experiment more with your “Light-wave Sensing and Communication” electronics kit in the future. Communication and sensing technologies are being developed by electrical and computer engineers at universities like Oklahoma State University.

What is in the Box?

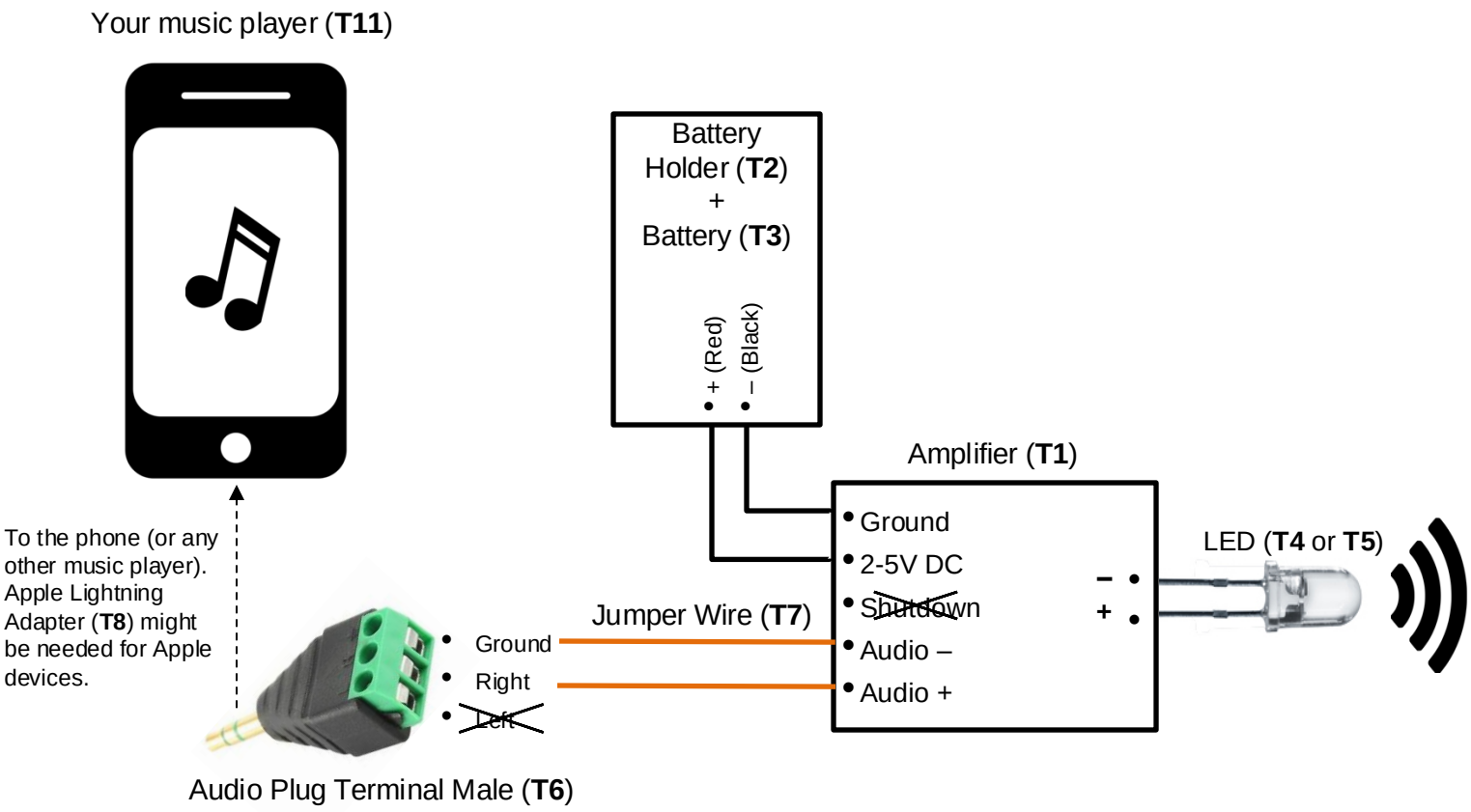


TRANSMITTER - LIST OF ITEMS

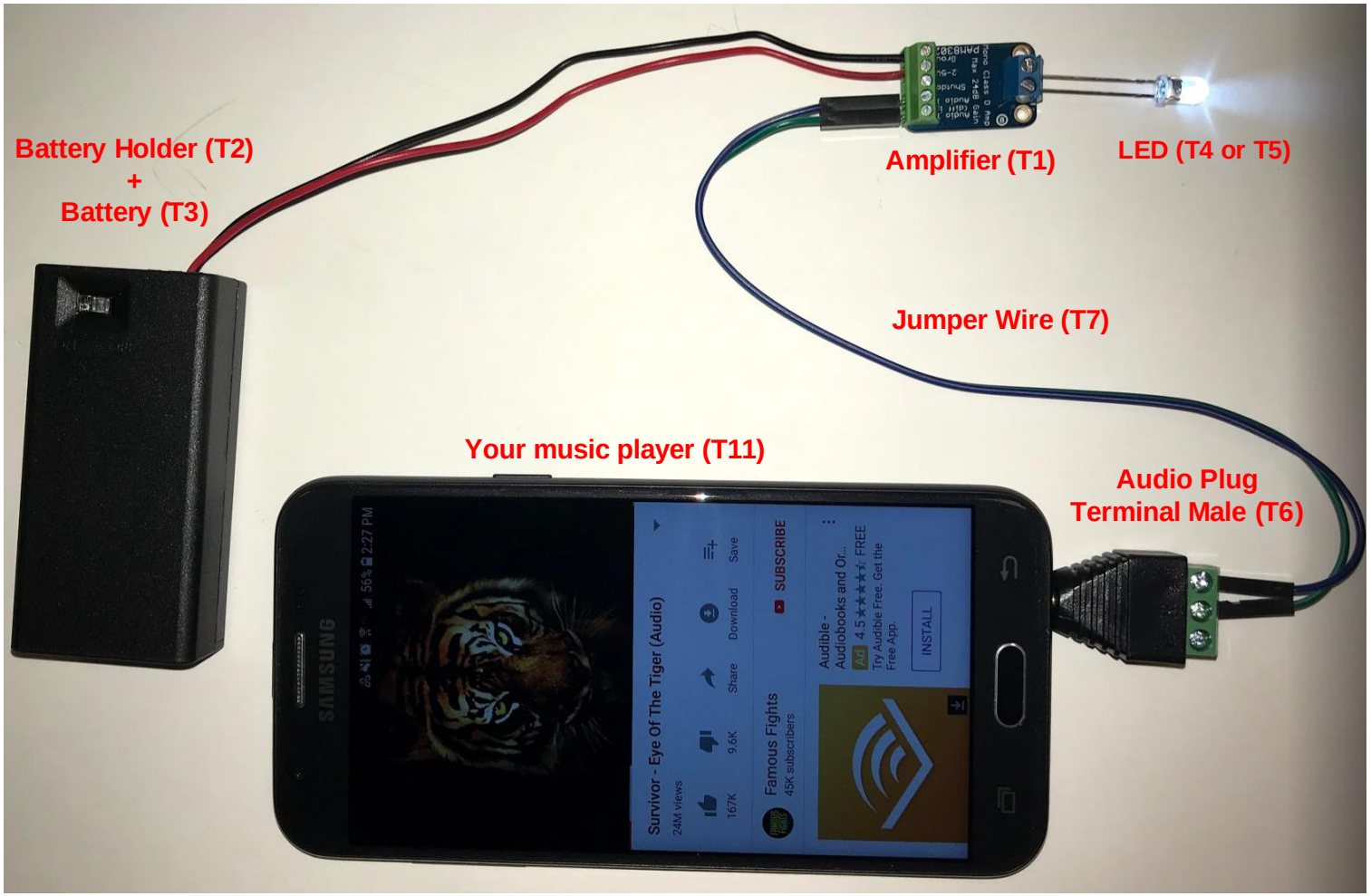
T#	Item	Image
T1	Amplifier	
T2	Battery Holder	
T3	Batteries (x2) - AA (will be provided once the all the connections is checked)	
T4	LED (White – Visible Light)	
T5	LED (Infra-Red – Invisible Light)	
T6	Audio Plug Terminal (Male)	
T7	Jumper Wire (Male-to-Male)	
T8	Apple Lightning Adapter (will be provided if needed)	
T9	Double Sided Adhesive White Foam Tape	
T10	Zip Tie	
T11	Your Music Player (e.g., phone, computer, etc.) (not included)	

TEXAS WIRELESS

TRANSMITTER - BLOCK DIAGRAM

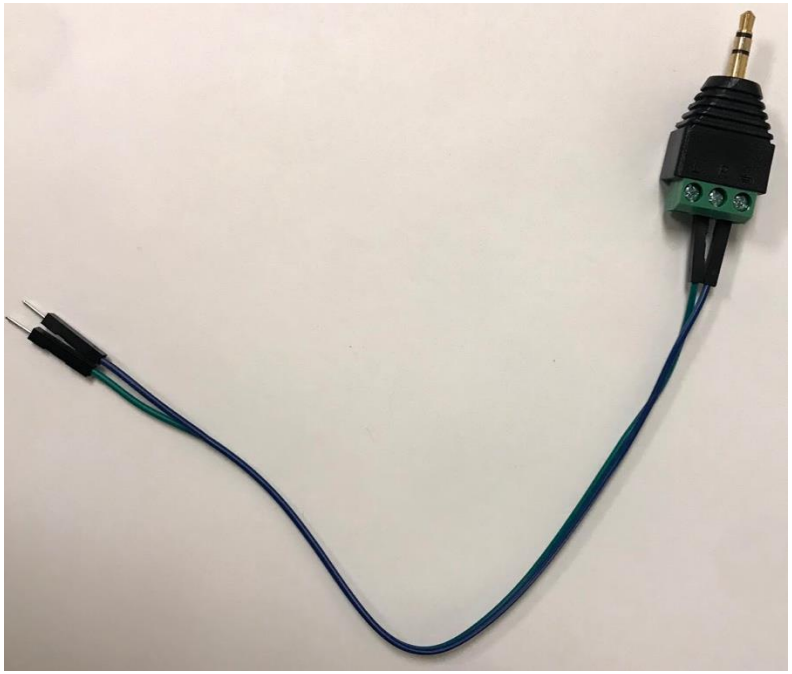


TRANSMITTER – REAL SETUP



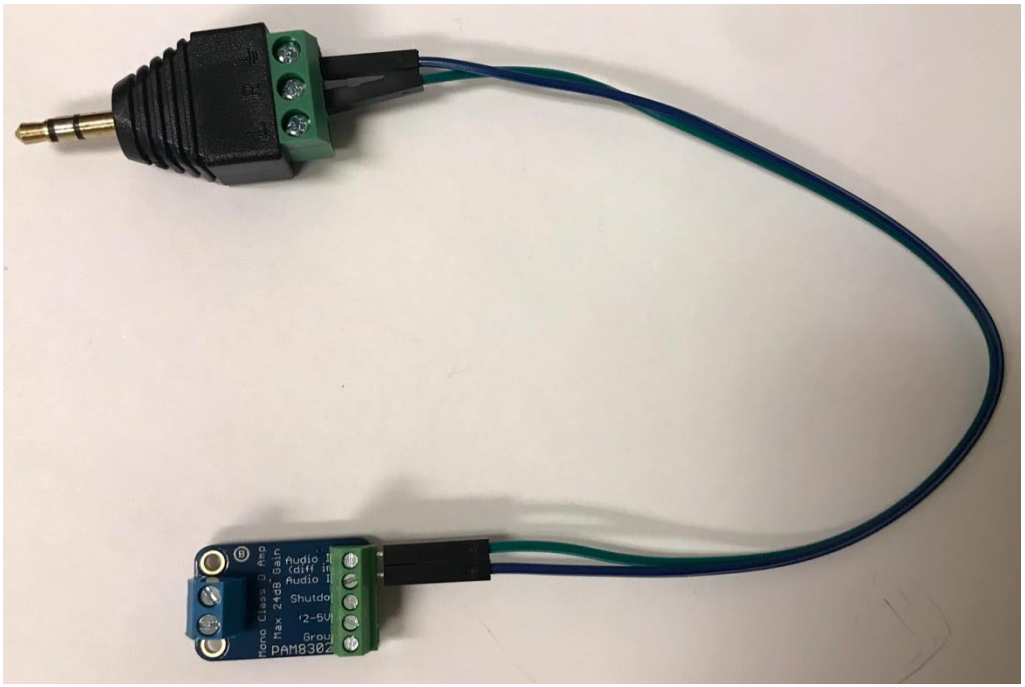
TRANSMITTER - STEPS

1. Connect *Jumper Wire (T7)* to *Audio Plug Terminal Male (T6).*



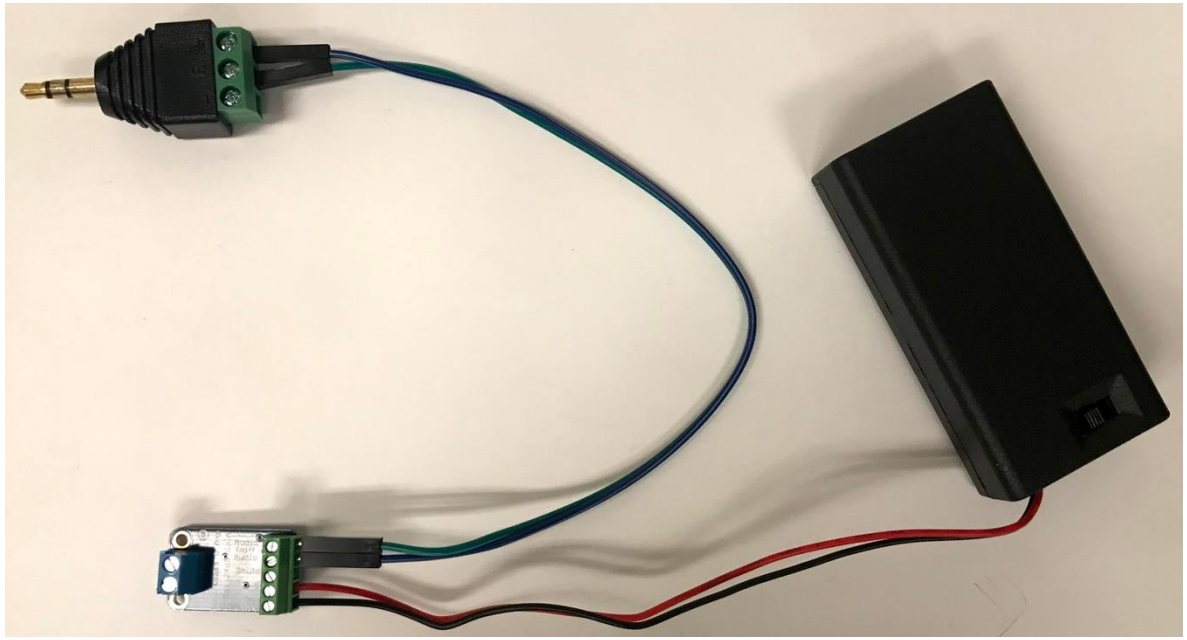
TRANSMITTER - STEPS

1. Connect *Jumper Wire (T7)* to *Audio Plug Terminal Male (T6).*
2. Connect the other end of the *Jumper Wire (T7)* to *Audio Amplifier (T1)*.



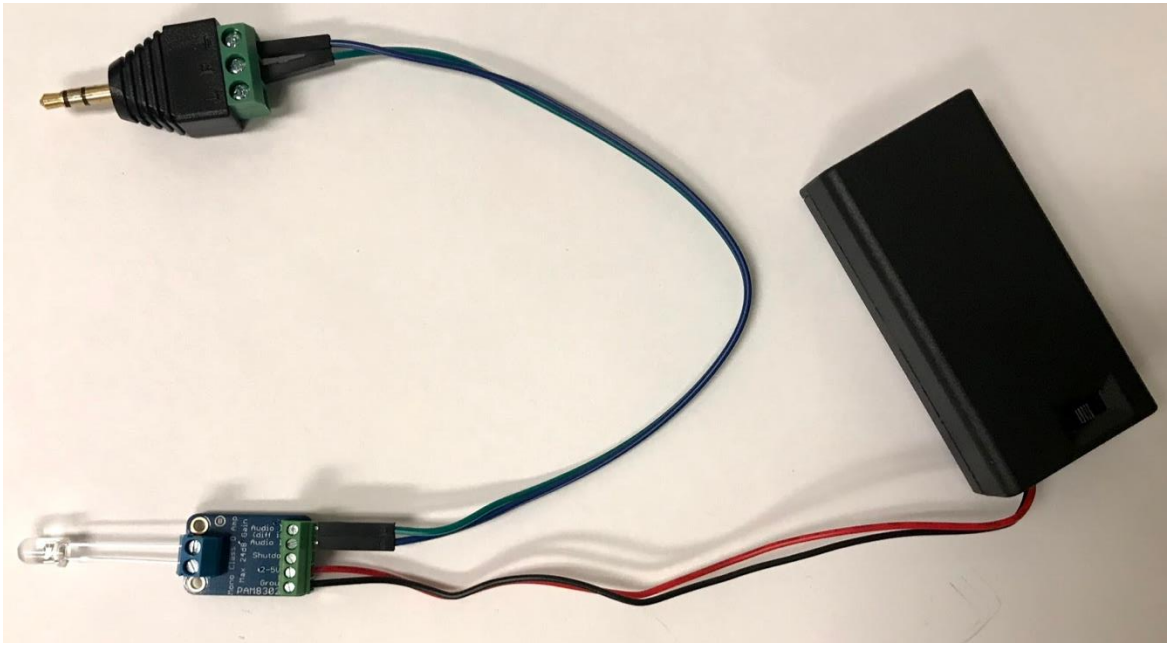
TRANSMITTER - STEPS

1. Connect *Jumper Wire (T7)* to *Audio Plug Terminal Male (T6).*
2. Connect the other end of the *Jumper Wire (T7)* to *Audio Amplifier (T1)*.
3. Connect *Battery Holder (T2)* to *Amplifier (T1)*. **Polarity must be correct.**



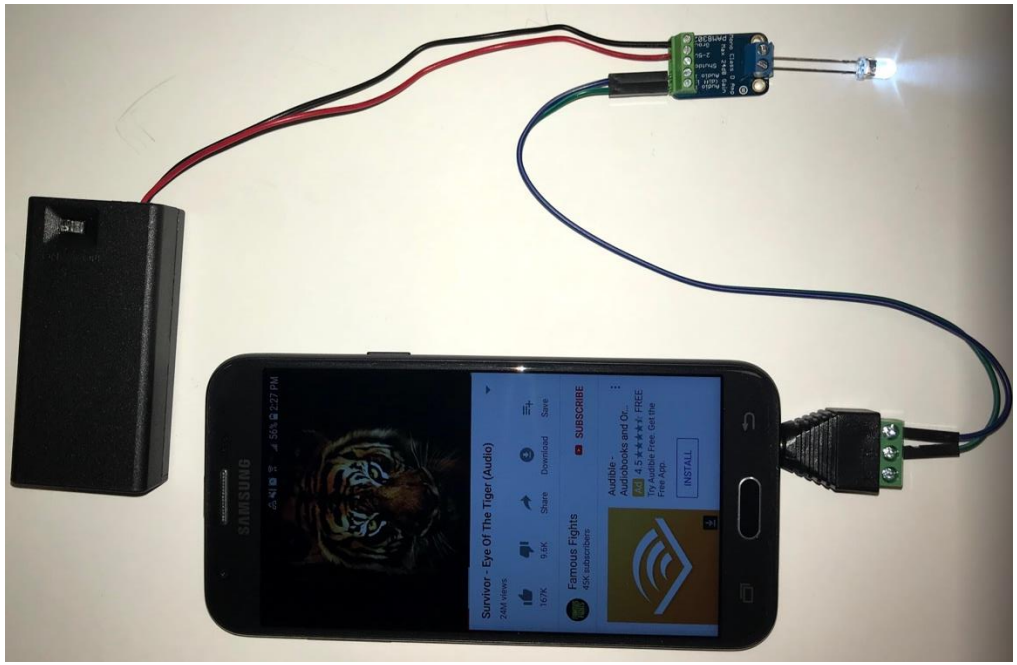
TRANSMITTER - STEPS

1. Connect *Jumper Wire (T7)* to *Audio Plug Terminal Male (T6).*
2. Connect the other end of the *Jumper Wire (T7)* to *Audio Amplifier (T1)*.
3. Connect *Battery Holder (T2)* to *Amplifier (T1)*. **Polarity must be correct.**
4. Connect *White LED (T4)* to *Amplifier (T1).*



TRANSMITTER - STEPS

1. Connect *Jumper Wire (T7)* to *Audio Plug Terminal Male (T6).*
2. Connect the other end of the *Jumper Wire (T7)* to *Audio Amplifier (T1)*.
3. Connect *Battery Holder (T2)* to *Amplifier (T1)*. **Polarity must be correct.**
4. Connect *White LED (T4)* to *Amplifier (T1).*
5. Connect *Audio Plug Terminal Male (T6)* to your music player (e.g., phone).



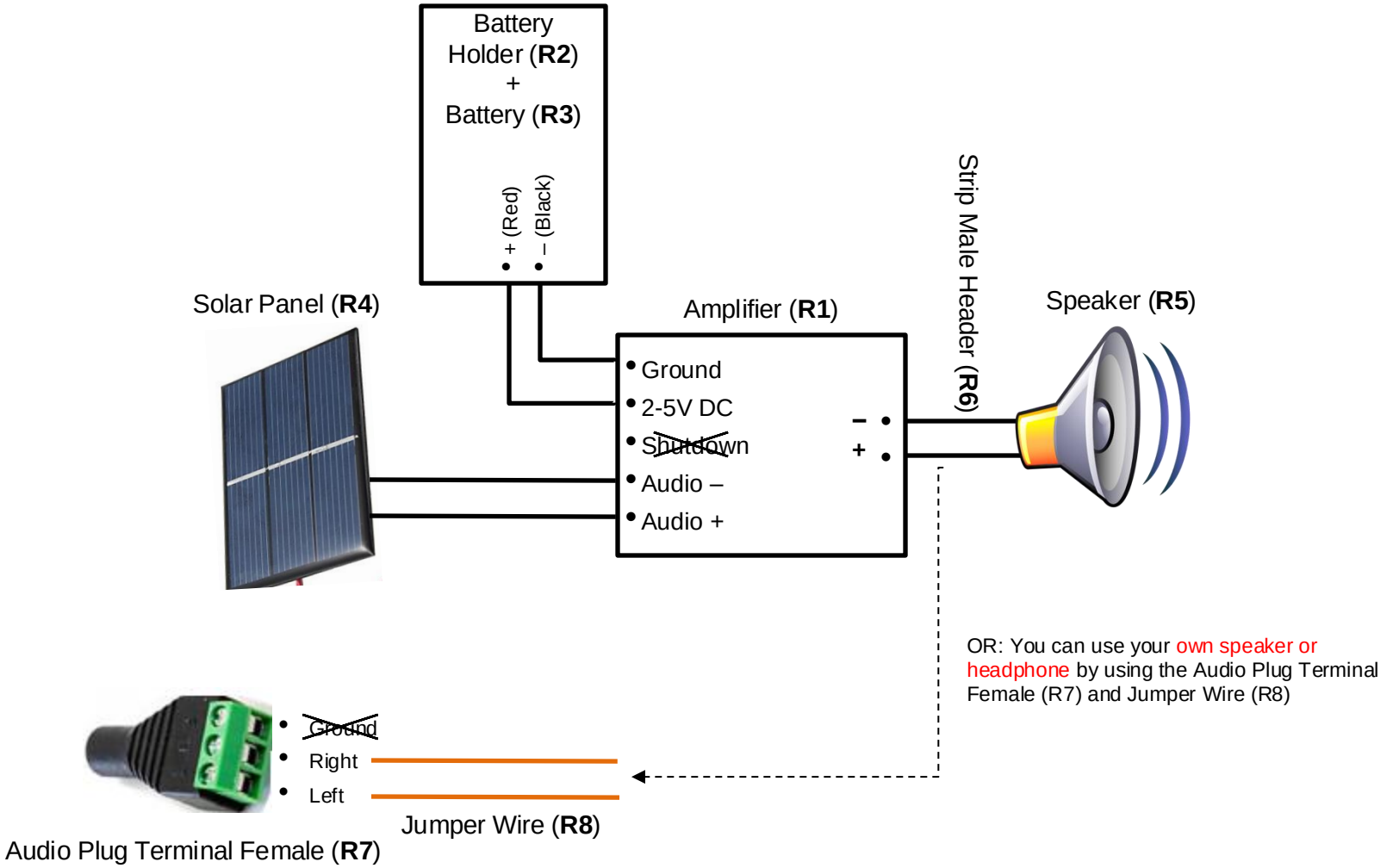
TRANSMITTER - STEPS

1. Connect *Jumper Wire (T7)* to *Audio Plug Terminal Male (T6).*
2. Connect the other end of the *Jumper Wire (T7)* to *Audio Amplifier (T1)*.
3. Connect *Battery Holder (T2)* to *Amplifier (T1)*. **Polarity must be correct.**
4. Connect *White LED (T4)* to *Amplifier (T1)*.
5. Connect *Audio Plug Terminal Male (T6)* to your music player (e.g., phone).
6. Your Transmitter setup is complete. You can now start putting your Receiver together.

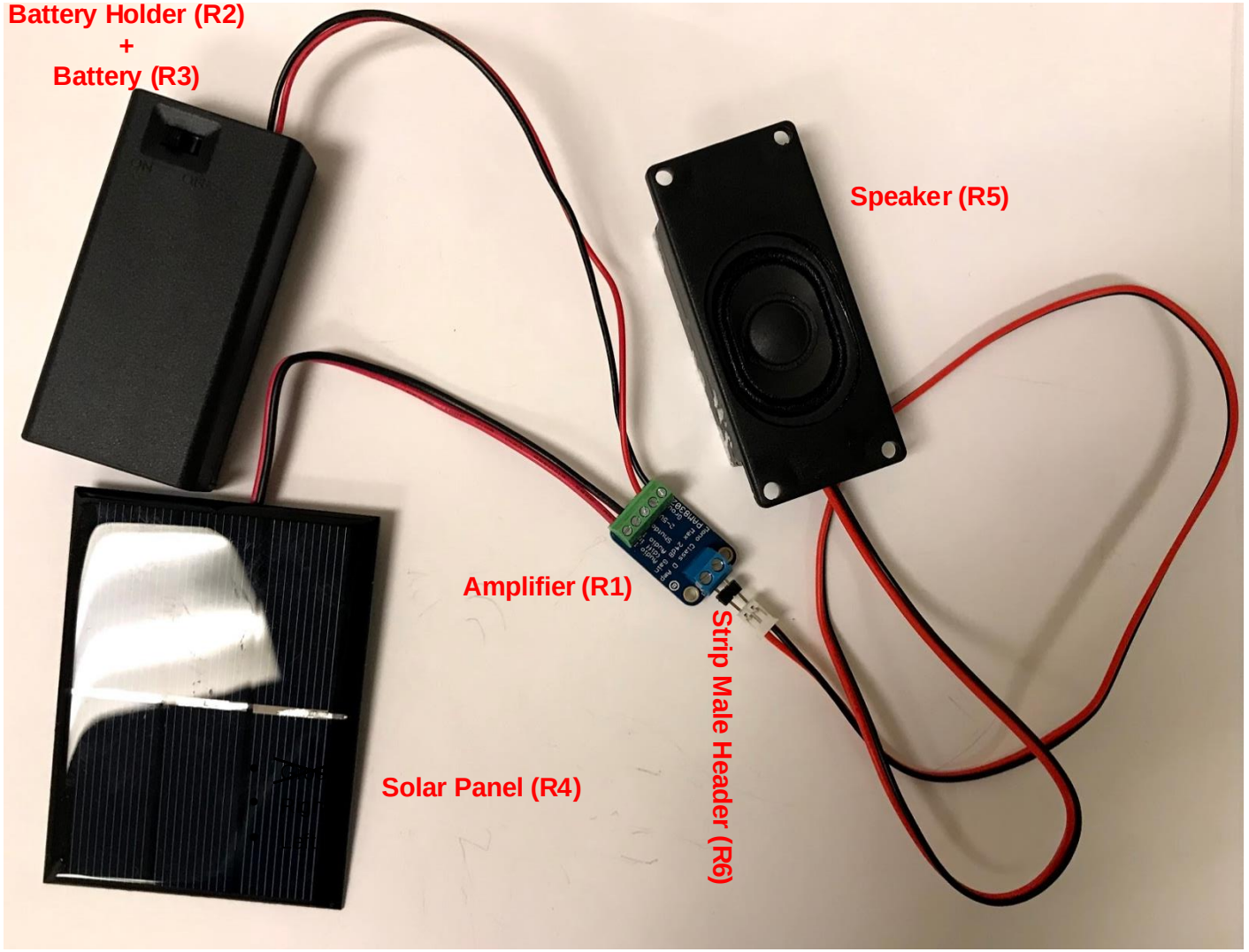
RECEIVER - LIST OF ITEMS

R#	Item	Image
R1	Amplifier	
R2	Battery Holder	
R3	Batteries (x2) - AA (will be provided once the all the connections is checked)	
R4	Solar Panel	
R5	Speaker	
R6	Strip Male Header	
R7	Audio Plug Terminal (Female)	
R8	Jumper Wire (Male-to-Male)	
R9	Double Sided Adhesive White Foam Tape (x2)	
R10	Zip Tie	

RECEIVER - BLOCK DIAGRAM

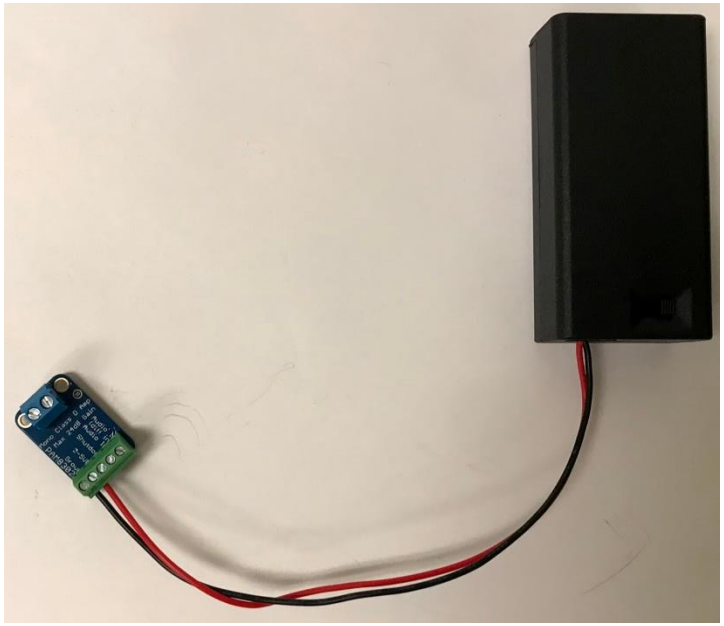


RECEIVER - REAL SETUP



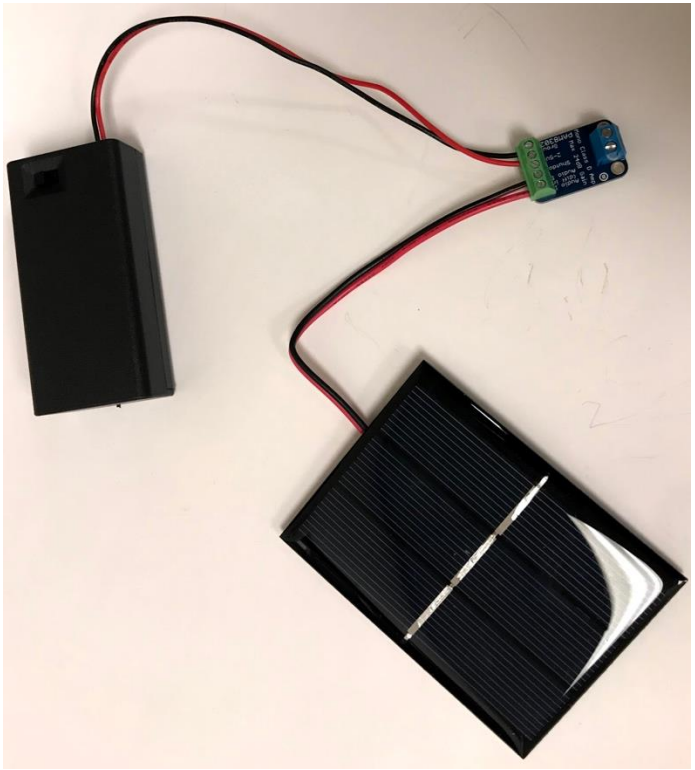
RECEIVER - STEPS

1. Connect *Battery Holder (R2)* to *Amplifier (R1)*. **Polarity must be correct.**



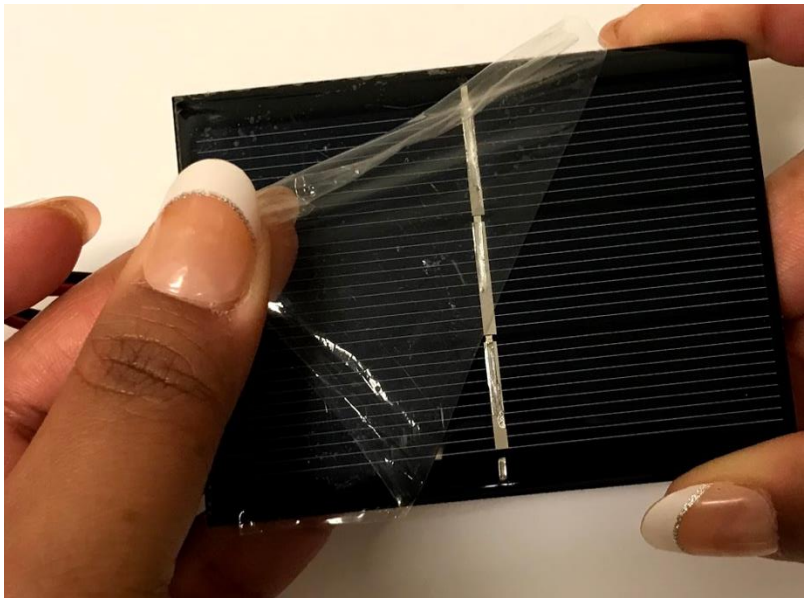
RECEIVER - STEPS

1. Connect *Battery Holder (R2)* to *Amplifier (R1)*. **Polarity must be correct.**
2. Connect *Solar Panel (R4)* to *Amplifier (R1)*.



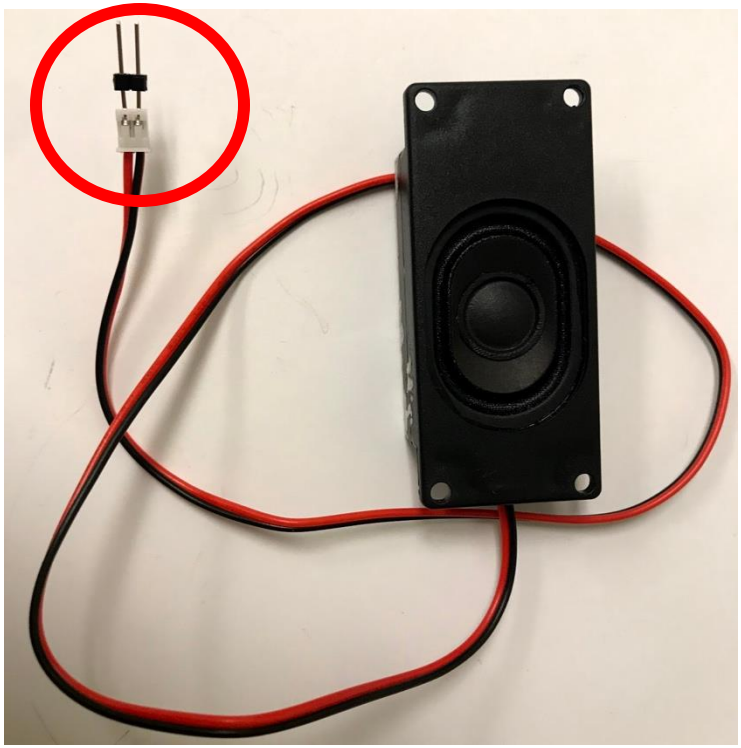
RECEIVER - STEPS

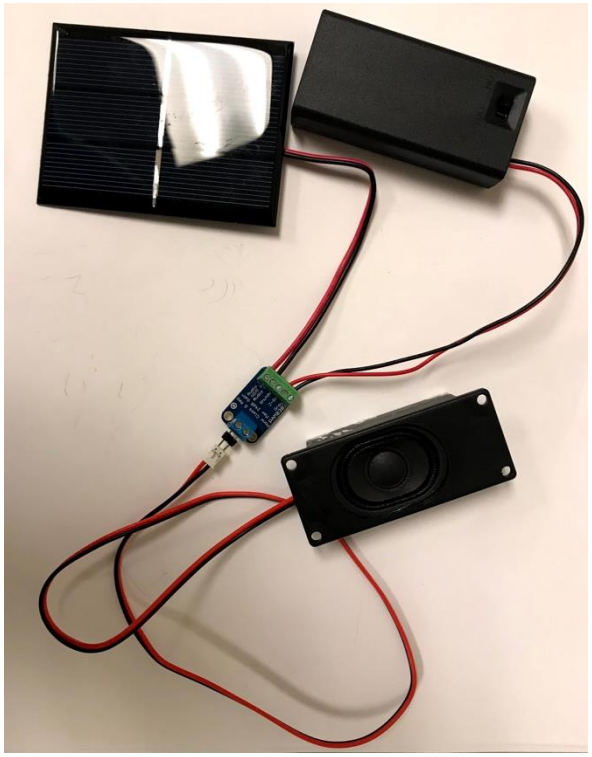
1. Connect *Battery Holder (R2)* to *Amplifier (R1)*. **Polarity must be correct.**
2. Connect *Solar Panel (R4)* to *Amplifier (R1)*.
3. Peel of plastic cover from *Solar Panel (R4)*.



RECEIVER - STEPS

1. Connect *Battery Holder (R2)* to *Amplifier (R1)*. **Polarity must be correct.**
2. Connect *Solar Panel (R4)* to *Amplifier (R1)*.
3. Peel of plastic cover from *Solar Panel (R4)*.
4. Connect *Speaker (R5)* to *Amplifier (R1)* by using *Strip Male Header (R6)*.





RECEIVER - STEPS

1. Connect *Battery Holder (R2)* to *Amplifier (R1)*. **Polarity must be correct.**
2. Connect *Solar Panel (R4)* to *Amplifier (R1)*.
3. Peel of plastic cover from *Solar Panel (R4)*.
4. Connect *Speaker (R5)* to *Amplifier (R1)* by using *Strip Male Header (R6)*.
5. Your Receiver is complete. You can now start Testing your system.

TESTING

Once both Transmitter and Receiver are ready. You can now test your Light-wave Sensing and Communication System.

1. Ask TA to check your connections,
   and then request *Batteries (T3, R3)* to put in *Battery Holders (T2, R2)*.
2. Turn ON the power switches for Transmitter and Receiver.
3. Play a song in your music player (e.g., phone).
4. Set the volume of your music player to the Max.
5. Point the LED towards the Solar Panel.
6. Can you hear the music from the speaker (at the Receiver)?
   - **a)** **Yes**. Then, enjoy the magic ☺. And do the followings:
     i. Observe the flicking of the light.
     ii. Try other LEDs as well: Infra-Red LED (T5) (**Invisible Light**) and color (Ask TA for color LEDs: Blue, Red, Green and Yellow LED). Does it still work? How does it compare?
     iii. Place a piece of paper between the light and solar panel. Can you still hear the music?
     iv. Invisible Infra-Red LED light can be **seen** by most of the android phone cameras. Try it yourself!
     v. Ask TA for TV a remote control. Press different buttons of the remote while holding towards the solar panel. What do you hear?
     vi. Ask TA for a mirror to experiment how reflected light can still transmit data. Test your system under sunlight (outside or next to a window). Does it still work?
   - **b)** **No**. Ask TA for help!

RECEIVER – STEPS – USING YOUR OWN SPEAKER

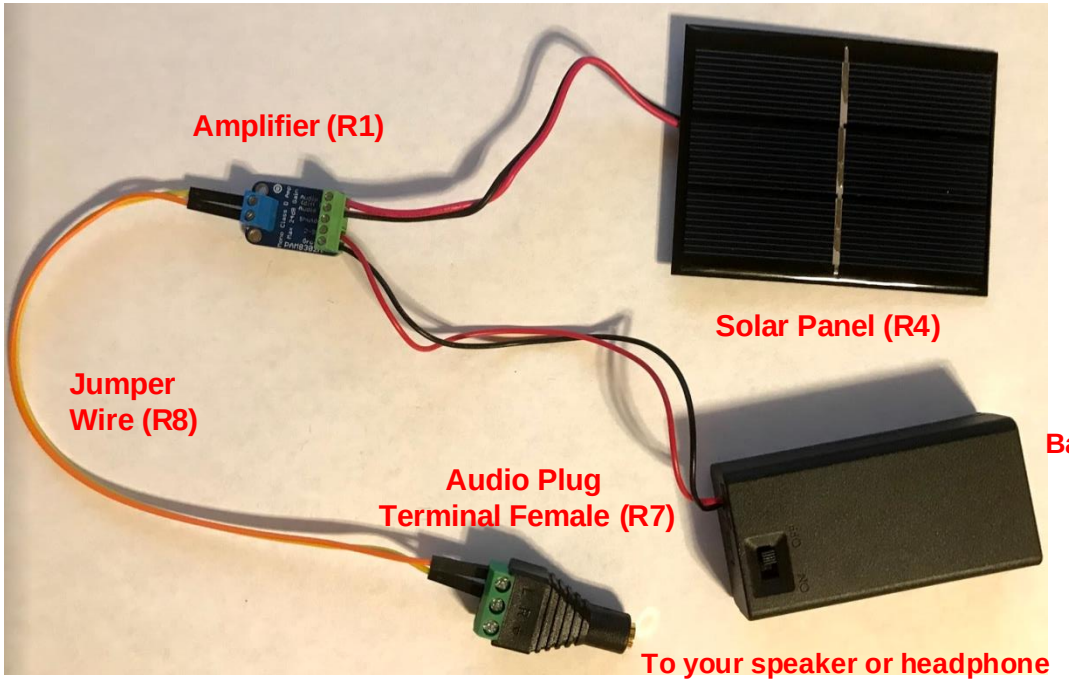
The speaker we provide is small and low-power. You can use your own speaker or headphone by doing the following steps:

a) Disconnect *Speaker (R5)* from *Amplifier (R1)*.

b) Connect *Jumper Wire (R7)* to *Audio Plug Terminal Female (R7).*

c) Connect the other end of *Jumper Wire (R7)* to *Amplifier (R1).*

d) Plug the *Female Audio Plug Terminal (R7)* to your own speaker or headphone.



FINAL TOUCHES

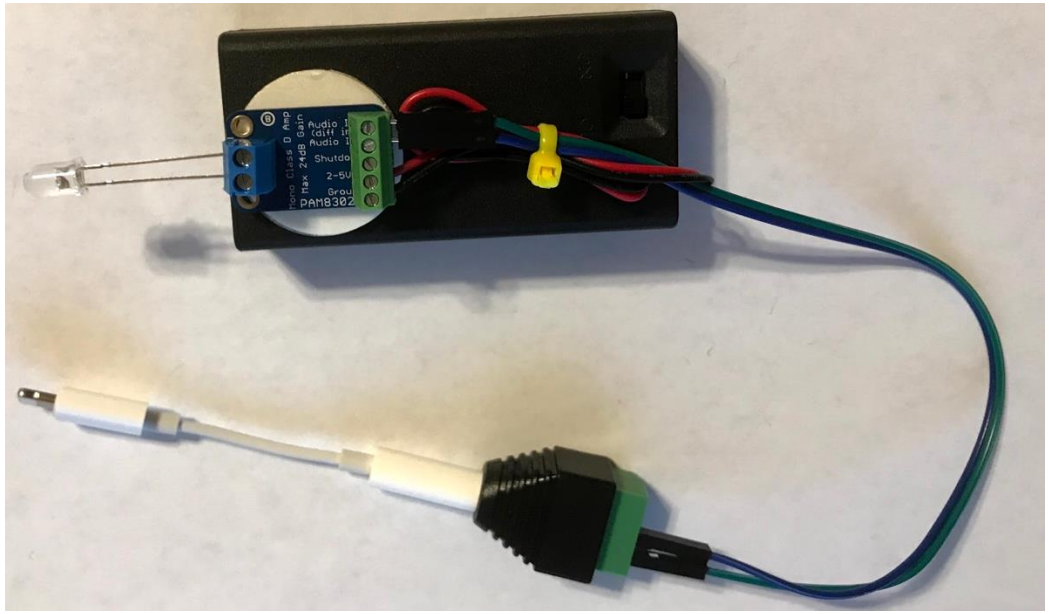
Once everything works as expected. You can now use “Double Sided Adhesive White Foam Tape” and “Zip Ties” to put your setup in a nice looking shape:

a) **Transmitter:** use adhesive tape to place Amplifier on Battery Holder.

b) **Receiver:** use adhesive tape to place Amplifier and Battery Holder on back side of Solar Panel.

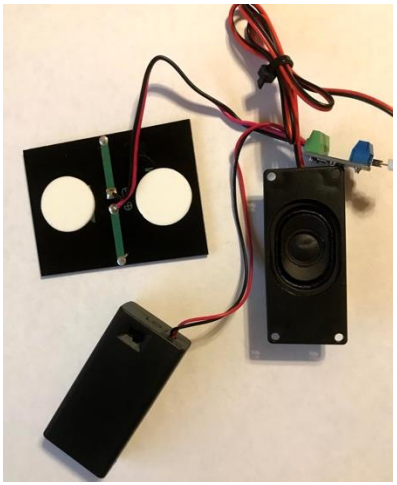
c) If needed, “Zip ties” can be used to make it more solid.

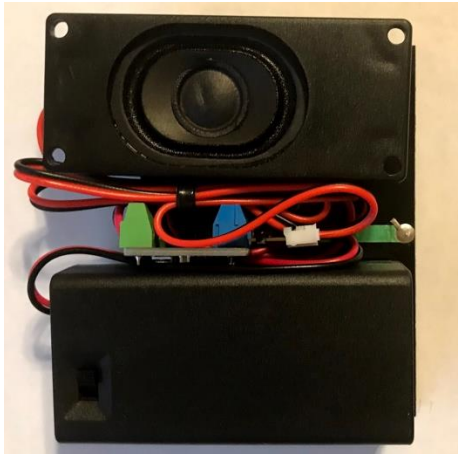
d) Here is what we did (see next slide also):



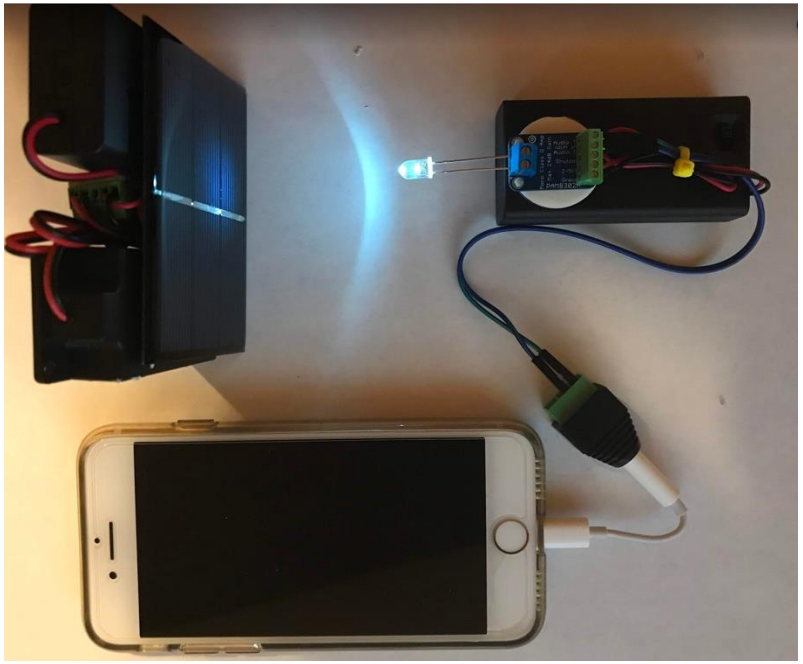
FINAL TOUCHES

e) And here is what we did for the receiver:









THANKS!

NOTES TO TAS

- Make sure questionnaire is completed.
- Make sure all amplifiers are set to **MAX** level (turned all the way to the clockwise end).
- Give other LED colors (Blue, Yellow, Red, Green) during TESTING.
- Give “Apple Lightning Adapter” to students who need it, and collect them back at the end of the session.
- Give mirrors to students to experiment the reflections.
- Make sure the music player is set to max volume.
- Make sure the plastic cover in solar panel is removed.
- Sunlight can impact the performance. It will increase noise level.
- Have a TV remote ready during testing.
- Have a scissor ready to cut the zip ties.
- Group photo time at the end ☺.